AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

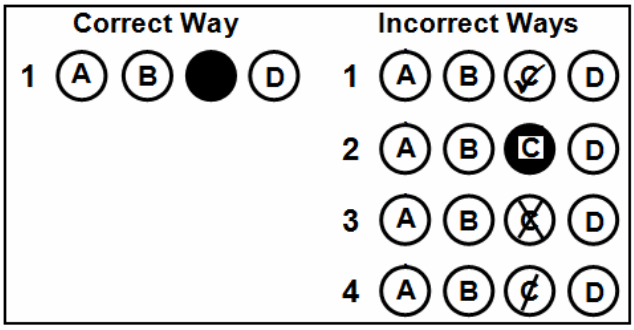
MAY 2014

Mathematics Paper I

Time: 40 minutes Marks: 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. The highest common factor (HCF) of $\frac{1}{x^2}$, $\frac{1}{xy}$ and $\frac{1}{xyz}$ is

   A. $\frac{1}{x}$
   B. $\frac{1}{y}$
   C. $\frac{1}{xy}$
   D. $\frac{1}{xyz}$

2. $\sqrt{(x+y)^2-4xy}$ is equal to

   A. $\pm(x-y)$
   B. $x-y$
   C. $\pm[(x+y)-2xy]$
   D. $(x+y)-2xy$

3. The highest common factor (HCF) of (x^2-9) and $(x+3)^2$ is

   A. $x+3$
   B. $x-3$
   C. $(x+3)^2(x-3)$
   D. $(x+3)^3(x-3)$

4. Which of the following is a proper fraction?

   A. $\frac{x^4}{x^3-1}$
   B. $\frac{x^2(x^2-1)}{x^3-1}$
   C. $\frac{x(x^2-1)}{x^4-1}$
   D. $\frac{x^5+5x}{x^4+4x}$

5. The solution set of $ax+a=a$ is

   A. $\{0\}$
   B. $\{2\}$
   C. $\{a\}$
   D. $\{2a\}$

6. The solution set of $\frac{\sqrt{x+2}}{2} = 0$ is equal to

   A. $\{2\}$
   B. $\{1\}$
   C. $\{-2\}$
   D. $\{-2, 2\}$

7. Which of the following equations has the solution set $\{-6, 2\}$?

   A. $|x+2| = 4$
   B. $|x-2| = 4$
   C. $|x+2| = -4$
   D. $|x-2| = -4$

8. If $a < b$ and $b < c$, then $a < c$ is in accordance with

   A. additive property.
   B. multiplicative property.
   C. transitive property.
   D. trichotomy.

9. For $x + 3y = 3$, the missing value of y in the table is

x	3	0
y	0	?

   A. -1
   B. 0
   C. 1
   D. 3

10. If $x - c = 0$ and $y - c = 0$, then their solution set is

    A. $\{(c, c)\}$
    B. $\{(-c, -c)\}$
    C. $\{(0, 0)\}$
    D. not possible

11. If (a, b) is a point in the cartesian plane and $a > 0$ and $b < 0$, then the point lies in the

    A. first quadrant.
    B. second quadrant.
    C. third quadrant.
    D. fourth quadrant.

12. The standard form of $\frac{x^2}{a}+\frac{x}{b}=\frac{c}{ab}$ is

A. $ax^2+bx+c=0$
B. $ax^2+bx-c=0$
C. $bx^2+ax+c=0$
D. $bx^2+ax-c=0$

13. The solution set of the quadratic equation $x^2+2x=0$ is

A. $\{-2\}$
B. $\{2\}$
C. $\{0,-2\}$
D. $\{0,2\}$

14. What should be added to make $x^2+\frac{3}{2}x$ a perfect square?

A. $\frac{3}{2}$
B. $\frac{3}{4}$
C. $\frac{9}{4}$
D. $\frac{9}{16}$

15. The distance between $(0,b)$ and $(-b,0)$ is

A. 0
B. $2b$
C. $\sqrt{2}\,b$
D. $2b^2$

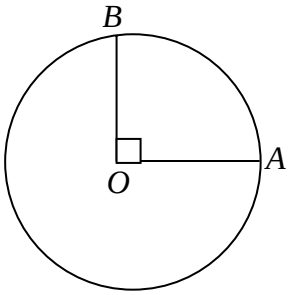
16. The point collinear to the points (0 , 0) and (1 , 1) is

A. (2 , 2)
B. (2 , 0)
C. (0 , 1)
D. (1 , – 1)

17. The distance between the points $(-3,0)$ and $(0,-1)$ is

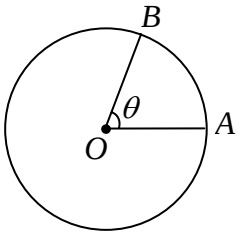
A. 8
B. $\sqrt{8}$
C. 10
D. $\sqrt{10}$

18. In the given diagram, $m\angle AOB$ is



A. $\frac{\pi}{4}$ radians.
B. 90 radians.
C. $\frac{\pi}{2}$ radians.
D. 180 radians.

19. In the given circle, lengths of arc AB and radius $\overline{OA}$ are equal. The measurement of angle θ is



A. 1 radian.
B. 2 radians.
C. π radians.
D. 2π radians.

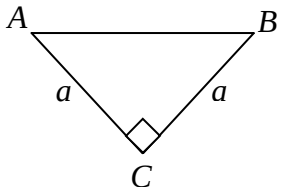
20. $\sin 30°$ is equal to

A. $\sin 60°$
B. $\cos 30°$
C. $\cos 60°$
D. $\text{cosec } 30°$

21. $\sqrt{2}\cos 45°$ is equal to

A. $\frac{1}{2}$
B. $\frac{1}{\sqrt{2}}$
C. 1
D. 2

22. In the given diagram, $m\overline{AB}$ is equal to

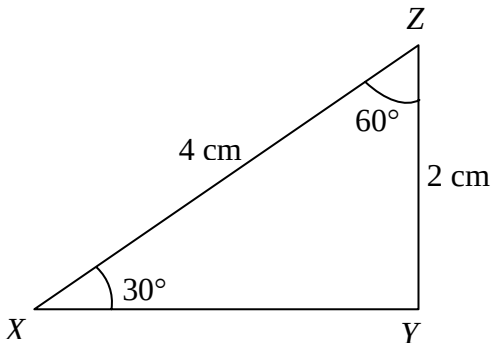


A. $2a^2$
B. $2a$
C. $\sqrt{2}\,a$
D. $\sqrt{2}\,a^2$

23. In the given ΔXYZ, $m\overline{XY}$ is equal to

A. $2cm$
B. $2\sqrt{3}cm$
C. $3cm$
D. $4cm$

NOT TO SCALE



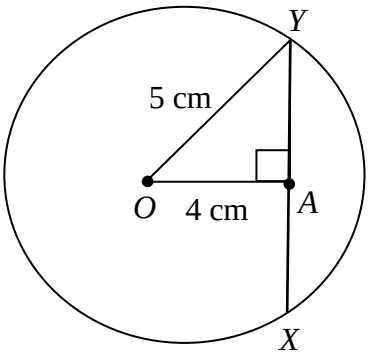
24. A rectangular table calendar with sides of length $25cm$ and $15cm$ has a diagonal length of

A. $27.15cm$
B. $28.15cm$
C. $29.15cm$
D. $30.15cm$

25. In the given circle, $\overline{OA}$ is perpendicular to $\overline{XY}$. If $m\overline{XY} = 6cm$, then $m\overline{AX}$ is equal to

A. $3cm$
B. $4cm$
C. $4.5cm$
D. $5cm$

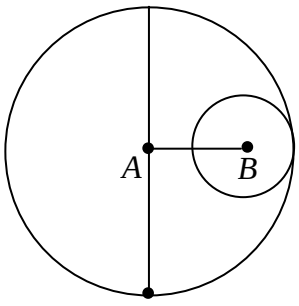
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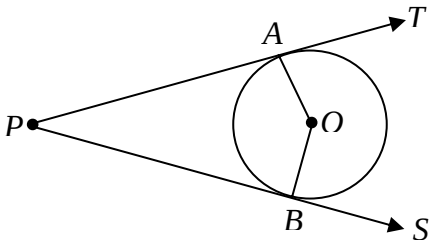
26. Two circles having centres A and B touch each other as shown in the figure. If the diameter of the circle with centre A is 24 cm and $m\overline{AB} = 9cm$, then the radius of the circle with centre B will be

A. $3cm$
B. $4cm$
C. $5cm$
D. $7cm$

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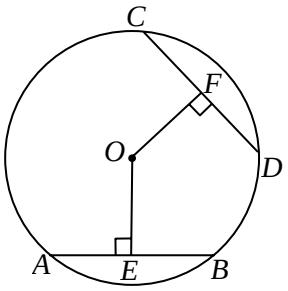
27. In the given figure, $\overrightarrow{PT}$ and $\overrightarrow{PS}$ are tangents to the circle at point *A* and point *B* respectively. Which of the following options is TRUE for the figure?



I. $m\overline{PA} = m\overline{PB}$
II. $\overline{OA} \cong \overline{OB}$
III. $\angle OAP \cong \angle OBP$

A. I only
B. I and II
C. II and III
D. I, II and III

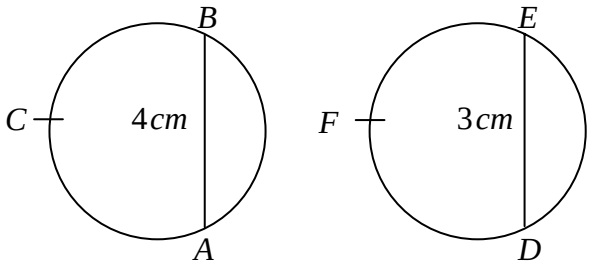
28. If *O* is the centre of the circle and $\overline{AB} \cong \overline{CD}$, then which of the following options is TRUE?



I. $\overline{OE} \cong \overline{OF}$
II. $\overline{AE} \cong \overline{EB}$
III. $\overline{CF} \cong \overline{EB}$

A. I, II and III
B. I only
C. II only
D. III only

29. For the given two congruent circles, consider the following statements.



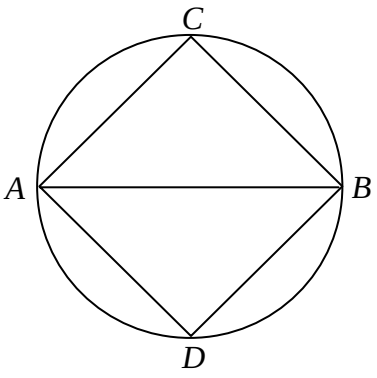
I. $m\overset{\frown}{AB} > m\overset{\frown}{ED}$
II. $m\overset{\frown}{ACB} > m\overset{\frown}{EFD}$
III. $m\overset{\frown}{ACB} < m\overset{\frown}{EFD}$

Which of the following options is CORRECT?

NOT TO SCALE

A. I only
B. II only
C. I and II
D. I and III

30. If $\overline{AB}$ is the diameter of the given circle, then $m\angle C + m\angle D$ is equal to



A. 60°
B. 90°
C. 120°
D. 180°

END OF PAPER

Please use this page for rough work